

Predicting Groundwater Levels in Southern New Mexico using Deep Learning

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1 Introduction

The western United States is increasingly facing severe water scarcity, threatening agricultural production, economic growth, and ecological stability. Declining in natural replenishment driven by climate change strains limited water supplies. In many places, this is exacerbated by rising consumption rates. In New Mexico, groundwater extraction has significantly outpaced aquifer recharge, causing significant declines in groundwater levels in many areas[2]. Increasing aridification is anticipated to cause large decreases in surface water availability in coming years, increasing demand for groundwater. This has been especially evident in the Rio Grande watershed this year. The current mega-drought and lack of snow pack in southern Colorado and Northern New Mexico, combined with large amounts of groundwater extraction led to the Rio Grande being entirely dry in Albuquerque for approximately three months[4]. In southern New Mexico, Elephant Butte Reservoir reached it's lowest level since 1971, and in July was at only 1.4 percent of capacity[6]. These hydrological stresses have contributed not only to interstate legal disputes, especially regarding compliance with the Rio Grande Compact, but also impacted agriculture and other economic areas, such as a chile harvest that is anticipated to be between twenty and fifty percent smaller than expected. [5].

Physically based hydrological models which simulate interactions between precipitation, temperature, evaporation, surface water, and groundwater require extensive resources and are difficult to update frequently. Data-driven approaches may be able to support more rapid assessment and continuous refinement of water-management strategies. This study explores the use of deep learning techniques to predict groundwater levels in southern New Mexico using long-term datasets of precipitation, temperature, streamflow, and groundwater levels, with the goal of evaluating whether a deep learning model can accurately reproduce hydrological patterns and deliver predictions suitable for guiding water management strategies.

2 Dataset

Precipitation and temperature records have been obtained from the National Oceanic and Atmospheric Administration (NOAA) station located just south of Las Cruces, New Mexico [8]. Streamflow data and groundwater level data have been obtained from the United States Geological Survey (USGS), for a stream gauge on the Rio Grande below Elephant Butte Dam[7] and twenty-three wells[1]. Precipitation and temperature datasets contain approximately 24,000 observations spanning 67 years, streamflow includes approximately 20,000 observations across 55 years, and groundwater well sampling frequency and temporal coverage varied widely. (Insert information about what format data is in.) We will select a common modeling period of at least twenty years to maximize overlapping coverage from all datasets, while retaining as much information as possible. Sources with significant quantities of missing data will be removed. Remaining missing values can be linearly interpolated to ensure continuity in data.

3 Proposed Methodology

To establish a baseline model, we propose using the Rio Grande Transboundary Integrated Hydrologic Model (RGTIHM, Hanson et al. (2024)), which is a physically based model derived from the numerical

solution of Poisson’s Equation in three-dimensional space. We propose comparing the RGTIHM model to a Recurrent Neural Network (RNN), most likely a Long Short-Term Memory (LSTM) model. RNNs have proven successful in regressive groundwater modeling [3], and LSTMs are well suited to regression problems involving temporal components. The main idea is to be able to compare the performance of a physically based model with a deep learning model against measured data.

4 Evaluation

The deep learning model (most likely LSTM) will be trained using a standard 70/30 training/test split, without shuffling the data in order to preserve seasonal and annual patterns. Because hydrologic variables often show strong temporal dependencies, we anticipate needing to time-lag some variables to provide the models with a means to learn seasonal cycles and long-term trends. The RGTIHM runs over the period 1940–2015 and is already calibrated; the challenge is to obtain outputs that can be evaluated against the deep learning model and against measured data. For this comparison, we will calculate the mean squared error, the mean absolute error, and the R-squared score between every model.

5 Expected Outcomes

We anticipate being able to predict groundwater levels with both models. However, we anticipate better performance from the LSTM, along with improved computational efficiency, measured in terms of time to perform computations and memory usage. We believe the most likely challenge will be a shortage of available groundwater data in the watershed of choice, due to differing sampling frequencies and periods across wells. Additionally, we are using a relatively small dataset, which may present challenges for effective training of the LSTM.

6 Timeline and milestones

- September 23: Create Github repository, upload all documents, datasets
- September 30: Finish data compilation, preprocessing, and initial exploration
- October 14: Complete XGBoost model
- October 28: Complete LSTM model
- November 6: Complete initial comparison of XGBoost, LSTM
- November 11: Complete any additional modifications required to the models after comparison, such as parameter tuning
- November 21: Complete written report
- December 2: Complete presentation

7 References

References

- [1]
- [2] Gretel Follingstad, PhD, Maurice Hall, PhD, Phil King, PhD, Adrian Oglesby, JD, Ramon Lucero, Leslie Hine, Larua Ziemer, Ladona Clayton, Jennifer Diffey, and Aron Balok. New Mexico 360 Groundwater Report. Technical report, New Mexico Groundwater Alliance, January 2026.
- [3] Jiangtao Liu, Chaopeng Shen, Fearghal O’Donncha, Yalan Song, Wei Zhi, Hylke E. Beck, Tadd Bindas, Nicholas Krabbel, and Kathryn Lawson. From rnns to transformers: benchmarking deep learning architectures for hydrologic prediction. *Hydrology and Earth System Sciences*, 2025.

- [4] Juana Summers. The Rio Grande is having its driest year on record, September 2026.
- [5] Leighanne Munoz. New mexico green chile supply drops amid heat. *Las Cruces Sun-News*, 2026.
- [6] NASA Earth Observatory and Lindsey Doermann. New mexico’s largest reservoir plunges to just 1.4 *Science Daily*, 2026.
- [7]
- [8]